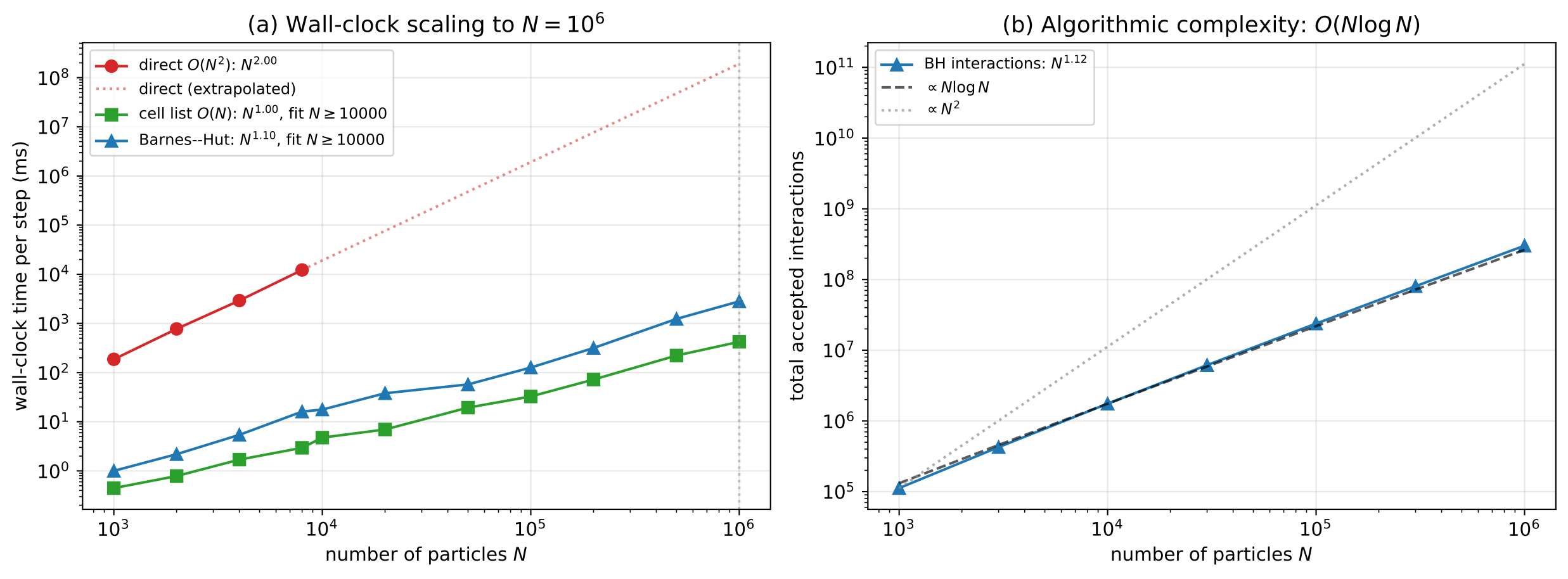
(a) Wall-clock scaling to $N = 10^6$
direct $O(N^2)$: $N^{2.00}$
direct (extrapolated)
cell list $O(N)$: $N^{1.00}$, fit $N \geq 10000$
Barnes--Hut: $N^{1.10}$, fit $N \geq 10000$
wall-clock time per step (ms)
number of particles N
(b) Algorithmic complexity: $O(N \log N)$
BH interactions: $N^{1.12}$
$\propto N \log N$
$\propto N^2$
total accepted interactions
number of particles N